

Graduate Algorithms

Quiz 1: Solutions

August 10, 2026

Sometimes in life the only trick is working hard at something for hours and hours.

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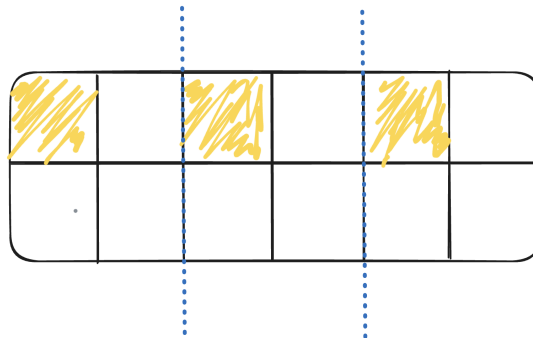
Question 1: Gold, Stone and Strategy

10 marks

Proposed by Arjun Maneesh Agarwal, modified from Indonesia LuMaT 2019, Final

King Midas and Medusa alternatively play on an 54×54 chessboard. Initially, all of the cells of the chessboard are made of wood. In each of his turn, Midas chooses three squares which are in the same column or row, and turns them into gold. In each of Medusa's turn, she can choose any 2 rows and 2 columns and turns the 4 intersecting cells into stone. King Midas wins if he can make a 2×2 grid square entirely of gold. Medusa wins if she can always prevent Midas doing so. Who wins? (Note: Midas can't turn stone to gold and not can Medusa turn gold to stone)

Solution. Consider this position:



Medusa can atmost affect two of the three 2×2 squares.

Thus, Midas can create 3 threats per turn (on large enough board) while Medusa only blocks 2.

So if we start with 27 threats (on the top 54×2 part of the board), Medusa would be able to difuse 18 of them.

On the remaining 9, we can add another square. Medusa will meanwhile will be able to block 6.

On the remaining 3, we can add yet another square. Medusa meanwhile will be able to block 2.

We complete the final square on this turn and win the game.

Notice, this strategy doesn't really depend on Medusa's moves.

Remark. The idea is really similar to one wins tic-tac-toe, by making two threats while the opponent can only block one. Also, 54 was chosen specifically for this reason.

Question 2: The Token Pile

10 marks

Proposed by Vardhan Kumar Ray

A pile of 3,141,592 tokens; Ada moves first. A move removes either one token, or exactly $\lfloor n/2 \rfloor$ tokens from a pile of size n . The player who leaves exactly one token wins. Who has a winning strategy?

Solution. Write the position simply as the number n of tokens in the pile, with the convention that the player *about to move* owns the position. From $n \geq 2$ the two legal moves lead to $n - 1$ and to $n - \lfloor n/2 \rfloor = \lceil n/2 \rceil$.

The game ends. Both successors are strictly smaller than n when $n \geq 2$, so the pile shrinks every turn; and from $n = 2$ both moves land on 1, so play terminates at exactly 1 and never overshoots.

Terminology. Call n an *N-position* if the player about to move can force a win, and a *P-position* otherwise, in which case the *previous* player, the one who just moved, wins. The position $n = 1$ is a P-position: whoever moved there has already won. For $n \geq 2$, the position n is N exactly when at least one of $n - 1$, $\lceil n/2 \rceil$ is P, and is P exactly when both are N.

Small cases. Working upwards from $n = 1$:

n	1	2	3	4	5	6	7	8	9	10	11	12
type	P	N	P	N	N	N	P	N	P	N	P	N

Every even entry is N, which suggests the following.

Claim. Every even $n \geq 2$ is an N-position.

Proof. Strong induction on k , where $n = 2k$.

Base case $k = 1$. From $n = 2$ both moves reach 1, a P-position, so 2 is an N-position.

Inductive step. Let $k \geq 2$ and suppose every even number smaller than $2k$ is an N-position. From $2k$ the two moves lead to $2k - 1$ and to $\lceil 2k/2 \rceil = k$.

- *If k is a P-position:* move to k . The opponent now faces a P-position, so we win. Hence $2k$ is N.
- *If k is an N-position:* move to $2k - 1$ instead. Since $k \geq 2$ we have $2k - 1 \geq 3$, so the opponent has not won. From $2k - 1$ the opponent's only options are $2k - 2$ and $\lceil (2k - 1)/2 \rceil = k$. Now $2k - 2$ is even and smaller than $2k$, so it is an N-position by the induction hypothesis; and k is an N-position by assumption. Every reply therefore hands *us* an N-position, from which we can force a win. Hence $2k$ is N. (Equivalently: $2k - 1$ is a P-position.)

Either way $2k$ is an N-position, completing the induction. ■

Conclusion. 3,141,592 is even, hence an N-position, so the player to move first wins: **Ada has a winning strategy.**

Question 3: Recurrences

10 marks

Proposed by Vardhan Kumar Ray

$T(n) = 5T(n/3) + \Theta(n)$, with $T(n) = \Theta(1)$ for $n \leq 3$. (a) Prove $T(n) = O(n^{\log_3 5})$ by substitution. (b) Verify with the Master theorem.

Solution. Throughout put $\alpha = \log_3 5$, so that $3^\alpha = 5$ and $\alpha \approx 1.465$. Fix constants $A > 0$ and $B > 0$ such that the driving term is at most An for $n > 3$, and $T(n) \leq B$ for $n \leq 3$. We ignore floors and ceilings (equivalently, take n to be a power of 3); this does not affect the asymptotics.

(a) Substitution method

The plain guess fails. Suppose we try $T(m) \leq cm^\alpha$ for all $m < n$. Then

$$T(n) \leq 5c \left(\frac{n}{3}\right)^\alpha + An = \frac{5}{3^\alpha} cn^\alpha + An = cn^\alpha + An,$$

using $3^\alpha = 5$. This exceeds cn^α by An , for *every* choice of c .

Strengthened hypothesis. Subtract a lower-order term. We claim there are constants $c, d > 0$ with

$$T(n) \leq cn^\alpha - dn \quad \text{for all } n \geq 1.$$

Inductive step. Let $n > 3$ and assume the bound for all smaller arguments. Then

$$T(n) \leq 5 \left[c \left(\frac{n}{3}\right)^\alpha - d \frac{n}{3} \right] + An = \frac{5}{3^\alpha} cn^\alpha - \frac{5}{3} dn + An = cn^\alpha - dn - \frac{2}{3} dn + An.$$

This is at most $cn^\alpha - dn$ precisely when $An \leq \frac{2}{3}dn$, that is, when

$$d \geq \frac{3A}{2}.$$

So take $d = 3A/2$. Note the strengthening is what makes this work: the extra $-\frac{2}{3}dn$ generated by the recursion is exactly the surplus that absorbs the additive An .

Base case. For $n \leq 3$ we have $T(n) \leq B$, and we need $B \leq cn^\alpha - dn$. The binding constraint is $n = 1$, where $n^\alpha = 1$ and the requirement reads $c - d \geq B$. (At $n = 2$ it reads $2^\alpha c - 2d \geq B$ with $2^\alpha \approx 2.76 > 2$, and at $n = 3$ it reads $5c - 3d \geq B$; both are weaker.) So

$$c := B + d = B + \frac{3A}{2}$$

suffices, and the induction goes through for all $n \geq 1$.

Conclusion. With $d = 3A/2$ and $c = B + 3A/2$,

$$T(n) \leq cn^\alpha - dn \leq cn^\alpha, \quad \text{so} \quad T(n) = O(n^{\log_3 5}).$$

(b) Master theorem

Matching against $T(n) = aT(n/b) + f(n)$ gives $a = 5$, $b = 3$ and $f(n) = \Theta(n)$. The watershed function is

$$n^{\log_b a} = n^{\log_3 5} = n^\alpha, \quad \alpha \approx 1.465.$$

Since $\alpha > 1$, set $\varepsilon = \alpha - 1 > 0$; then $f(n) = \Theta(n) = O(n^{\alpha-\varepsilon})$, so f is polynomially smaller than the watershed and **Case 1** applies:

$$T(n) = \Theta(n^{\log_3 5}).$$

This agrees with part (a), and is in fact slightly stronger, the Master theorem delivers a matching lower bound, whereas the substitution argument above only established the upper bound.

Question 4: Counting Triangles

10 marks

Proposed by Arjun Maneesh Agarwal

We have an algorithm (refer to **algorithm 1**) to count the number of triangles in a graph. Could you calculate the running time for the given algorithm, assuming checking if an edge exists is $O(1)$?

Algorithm 1 An algorithm to count the number of triangles in a graph

```
1: count  $\leftarrow$  0
2: for each vertex  $v \in V$  do
3:   for each pair  $u, w \in N(v)$  such that  $u \neq w$  do
4:     if  $uw \in E$  then
5:       count  $\leftarrow$  count + 1
6:     end if
7:   end for
8: end for
9: return count/3
```

Solution. The first loop goes over all the vertices $v \in V$. For every vertex, it considers all pairs in its neighbors that is $\binom{\deg(v)}{2} = O(\deg(v)^2)$. Thus, the total run time is

$$O\left(\sum_{v \in V} \deg(v)^2\right).$$

Remark. The question asked for running time. We never asked for worst case time complexity. We don't know why people were so eager to give us worst case complexity.